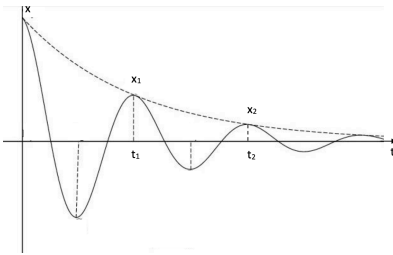
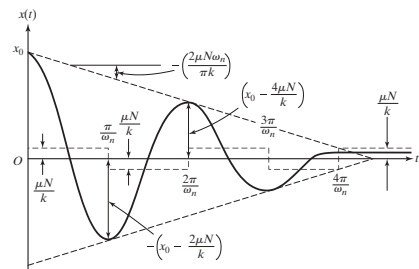


Mass <u>Translational</u> $\sum F = ma = m \frac{v(t)}{dt} = m \frac{d^2x(t)}{dt^2}$ <u>Rotational</u> $\sum T = J\alpha = \frac{Jd\Omega(t)}{dt} = J \frac{d^2\theta(t)}{dt^2}$	Spring Opposes Displacement <u>Translational</u> $F_s = kx$ <u>Rotational</u> $T_s = \mathcal{K}\theta$	Damper Opposes Velocity $\zeta < \frac{1}{\sqrt{2}}$ <u>Translational</u> $F_d = c\dot{x} \quad F = \mu_k N$ <u>Rotational</u> $T_d = C\dot{\theta}$	Equivalent Spring Constants Springs in parallel: $k_{eq} = k_1 + k_2$ Springs in series: $\frac{1}{k_{eq}} = \frac{1}{k_1} + \frac{1}{k_2} + \frac{1}{k_3} + \dots$ Linear to rotational $x = r\theta \quad \dot{x} = r\dot{\theta} \quad \ddot{x} = r\ddot{\theta}$
Harmonic Motion $\theta = \omega t$ $\omega = \frac{\theta}{t} = \frac{2\pi}{t} = 2\pi f$ $\sin(\omega t) = \cos\left(\omega t - \frac{\pi}{2}\right)$ $-\sin(\omega t) = \sin(\omega t + 180^\circ)$	Energy <u>Translational KE</u> $T = \frac{1}{2}m\dot{x}^2$ <u>Rotational KE</u> $T = \frac{1}{2}J\dot{\theta}^2$ <u>Spring PE</u> $U = \frac{1}{2}k(\Delta x)^2$	Fourier Expansion $y(t) = \frac{a_0}{2} + \sum_{i=1}^N a_n \cos\left(\frac{2n\pi t_i}{\tau}\right) + b_n \sin\left(\frac{2n\pi t_i}{\tau}\right)$ $a_0 = \frac{2}{N} \sum_{i=1}^N x_i$ $a_n = \frac{2}{N} \sum_{i=1}^N x_i \cos\left(\frac{2n\pi t_i}{\tau}\right)$ $b_n = \frac{2}{N} \sum_{i=1}^N x_i \sin\left(\frac{2n\pi t_i}{\tau}\right)$	Vibrations <u>Natural Frequency</u> $\omega_n = \sqrt{\frac{k_{eq}}{m_{eq}}} = \sqrt{\frac{g}{\delta_{st}}}$ <u>Static Equilibrium</u> $k = \frac{mg}{\delta_{st}}$ <u>Axial Stiffness</u> $k = \frac{AE}{L}$
Standard Mass-Spring ODE $m\ddot{x} + kx = 0$ <u>Standard Form</u> $\ddot{x} + \frac{k}{m}x = 0 \quad \ddot{x} + \omega_n^2 x = 0$ <u>Solution</u> $x(t) = x_0 \cos(\omega_n t) + \frac{\dot{x}_0}{\omega_n} \sin(\omega_n t)$	Standard Mass-Spring-Damper ODE $m\ddot{x} + c\dot{x} + kx = 0$ <u>Standard Form</u> $\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2 x = 0$ $\zeta = \frac{c}{2m\omega_n}$ <u>Underdamped</u>: $0 < \zeta < 1$ Roots: $s = -\zeta\omega_n \pm i\omega_d$ $\omega_d = \omega_n\sqrt{1 - \zeta^2}$ $x(t) = e^{-\zeta\omega_n t} \left[x_0 \cos(\omega_d t) + \frac{\dot{x}_0 + \zeta\omega_n x_0}{\omega_d} \sin(\omega_d t) \right]$ <u>Critically Damped</u>: $\zeta = 1$ Roots: $s = -\omega_n$ $x(t) = [x_0 + (\dot{x}_0 + \omega_n x_0)t]e^{-\omega_n t}$ <u>Overdamped</u>: $\zeta > 1$ Roots: $s_{1,2} = -\zeta\omega_n \pm \sqrt{\zeta^2 - 1}$ $x(t) = C_1 e^{s_1 t} + C_2 e^{s_2 t}$ $C_1 = \frac{x_0\omega_n(\zeta + \sqrt{\zeta^2 - 1}) + \dot{x}_0}{2\omega_n\sqrt{\zeta^2 - 1}}$ $C_2 = \frac{-x_0\omega_n(\zeta - \sqrt{\zeta^2 - 1}) - \dot{x}_0}{2\omega_n\sqrt{\zeta^2 - 1}}$	Coulomb damping $m\ddot{x} + kx = \pm\mu N$ <u>Standard Form</u> $\ddot{x} + \omega_n^2 x = \pm \frac{\mu N}{m}$ <u>Solution for +</u> $x(t) = \left(x_0 - \frac{\mu N}{k}\right) \cos(\omega_n t) + \frac{\mu N}{k}$ <u>Solution for -</u> $x(t) = \left(x_0 - \frac{3\mu N}{k}\right) \cos(\omega_n t) - \frac{\mu N}{k}$ Where $\tau_d = \frac{2\pi}{\omega_n} \quad \omega_n = \omega_d$ $x(\tau_d) = x_0 - \frac{4\mu N}{k}$ r is the # of $\frac{1}{2}$ cycles $r \leq \frac{x_0 - \frac{\mu N}{k}}{\frac{2\mu N}{k}}$	
Logarithmic Decrement  The rate at which the amplitude of vibration is decreasing. x_1 and x_2 are successive amplitudes one period apart. $\delta = \ln\left(\frac{x_1}{x_2}\right) = \zeta\omega_n\tau_d \quad \tau_d = t_2 - t_1$ $\delta = \frac{2\pi\zeta}{\sqrt{1 - \zeta^2}} \quad \zeta = \frac{\delta}{\sqrt{(2\pi)^2 + \delta^2}}$ $\tau_d = \frac{2\pi}{\omega_d} = \frac{2\pi}{\sqrt{1 - \zeta^2}\omega_n}$ $\delta = \frac{1}{n} \ln\left(\frac{x_1}{x_{n+1}}\right)$ Where n is the # of cycles			

Harmonically Forced with No Damping

$$m\ddot{x} + kx = F_0 \cos(\omega t)$$

The solution is

$$x(t) = \left(x_0 - \frac{F_0}{k - m\omega^2}\right) \cos(\omega_n t) + \left(\frac{\dot{x}_0}{\omega_n}\right) \sin(\omega_n t) + \frac{F_0}{k - m\omega^2} \cos(\omega t) \leftarrow x_p$$

Forced vibration with no damping with sin

$$m\ddot{x} + kx = F_0 \sin(\omega t)$$

Solution

$$x(t) = x_0 \cos(\omega_n t) + \left(\frac{\dot{x}_0 - \frac{F_0}{k - m\omega^2}}{\omega_n}\right) \sin(\omega_n t) + \frac{F_0}{k - m\omega^2} \sin(\omega t) \leftarrow x_p$$

The maximum value of the forced response is given by the constant:

$$X_{max} = \frac{F_0}{|k - m\omega^2|} = \frac{\delta_{st}}{1 - \left(\frac{\omega}{\omega_n}\right)^2}$$

Static deflection is

$$F_0 = k \cdot \delta_{st}$$

Magnification factor or Amplitude Ratio

$$\frac{X_{max}}{\delta_{st}} = \frac{1}{1 - \left(\frac{\omega}{\omega_n}\right)^2}$$

The frequency ratio is defined by

$$r = \frac{\omega}{\omega_n}$$

Harmonically Forced with Damping

$$m\ddot{x} + c\dot{x} + kx = F_0 \cos(\omega t)$$

Solution

$$x(t) = X_0 e^{-\zeta \omega_n t} \cos(\omega_d t - \phi_0) + X \cos(\omega t - \phi) \leftarrow x_p$$

$$X_0 = \left[\left(x_0 - X \cos(\phi)\right)^2 + \frac{1}{\omega_d^2} \left(\zeta \omega_n x_0 + \dot{x}_0 - \zeta \omega_n X \cos(\phi) - \omega X \sin(\phi)\right)^2 \right]^{\frac{1}{2}}$$

$$\tan(\phi_0) = \frac{\zeta \omega_n x_0 + \dot{x}_0 - \zeta \omega_n X \cos(\phi) - \omega X \sin(\phi)}{\omega_d (x_0 - X \cos(\phi))}$$

Steady-state response amplitude

$$X = \frac{F_0}{\sqrt{(k - m\omega^2)^2 + c^2 \omega^2}}$$

Phase angle

$$\tan(\phi) = \frac{c\omega}{k - m\omega^2}$$

Resonance and magnification

$$A_{magnification} = \frac{X_{max}}{\delta_{st}} = \frac{1}{\sqrt{(1 - r^2)^2 + (2\zeta r)^2}}$$

$$\phi = \tan^{-1} \left[\frac{2\zeta \frac{\omega}{\omega_n}}{1 - \left(\frac{\omega}{\omega_n}\right)^2} \right] = \tan^{-1} \left[\frac{2\zeta r}{1 - r^2} \right]$$

$$\delta_{st} = \frac{F_0}{k_{eq}} \quad r = \frac{\omega}{\omega_n}$$

$$M_{max} \text{ is at } r = \sqrt{1 - 2\zeta^2}$$

$$X_{max} = \frac{\delta_{st}}{2\zeta \sqrt{1 - \zeta^2}}$$

Complex Forced with Damping

$$m\ddot{x} + c\dot{x} + kx = f_0 e^{i\omega t}$$

Steady State Solution

$$x_p(t) = X e^{i(\omega t - \phi)}$$

$$\tan(\phi) = \frac{c\omega}{k - m\omega^2}$$

$$X = \frac{F_0}{\sqrt{(k - m\omega^2)^2 + (c\omega)^2}}$$

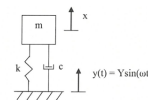
If Forcing is $F_0 \cos(\omega t) \rightarrow$ real part of solution

If Forcing is $F_0 \sin(\omega t) \rightarrow$ imaginary part of solution

Base Citation

$$m\ddot{x} + c\dot{x} + kx = c\dot{y} + ky$$

$$y(t) = Y \sin(\omega t) \quad \dot{y}(t) = Y \omega \cos(\omega t)$$



Solution

$$x_p(t) = X \sin(\omega t - \phi)$$

$$\frac{X}{Y} = \left[\frac{k^2 + (c\omega)^2}{(k - m\omega^2)^2 + (c\omega)^2} \right]^{\frac{1}{2}} = \left[\frac{1 + (2\zeta r)^2}{(1 - r^2)^2 + (2\zeta r)^2} \right]^{\frac{1}{2}}$$

$$\phi = \tan^{-1} \left[\frac{mc\omega^3}{k(k - m\omega^2) + (c\omega)^2} \right] = \tan^{-1} \left[\frac{2\zeta r^3}{1 + (4\zeta^2 - 1)r^2} \right]$$

$$M_{max} = \left(\frac{X}{\delta_{st}} \right)_{max} = \frac{1}{2\zeta \sqrt{1 - \zeta^2}}$$

$$M_{max} = \left(\frac{\theta}{\theta_{st}} \right)_{max} = \frac{1}{2\zeta \sqrt{1 - \zeta^2}}$$

Remember Logan – It's Just an Exam

Philippians 4:6-7:

“Do not be anxious about anything, but in everything by prayer and supplication with thanksgiving let your requests be made known to God.

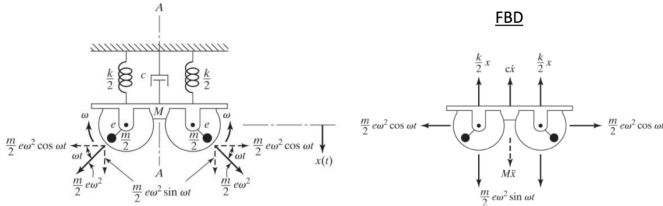
And the peace of God, which surpasses all understanding, will guard your hearts and your minds in Christ Jesus.”

“Equilibrium is temporary - keep pushing forward.”

“Displacement doesn't define you - acceleration does.”

“No damping can stop determination.”

Unbalanced Rotation



FBD will be

$$M\ddot{x} + c\dot{x} + kx = me\omega^2 \sin(\omega t)$$

Solution:

$$x_p(t) = X \sin(\omega t - \phi)$$

$$X = \frac{me\omega^2}{\sqrt{(k - M\omega^2)^2 + (c\omega)^2}}$$

$$\phi = \tan^{-1} \left(\frac{c\omega}{k - M\omega^2} \right)$$

Max amplifier becomes

$$\frac{MX}{me} = \frac{r^2}{[(1 - r^2)^2 + (2\zeta r)^2]^{\frac{1}{2}}}$$

$$\phi = \tan^{-1} \left[\frac{2\zeta r}{1 - r^2} \right]$$

If force exerted on the ground to be:

$$F(t) = kx(t) + c\dot{x}(t)$$

Then the magnitude of F is given by:

$$|F| = me\omega^2 \left[\frac{1 + 4\zeta^2 r^2}{(1 - r^2)^2 + 4\zeta^2 r^2} \right]^{\frac{1}{2}}$$

General Forcing Functions

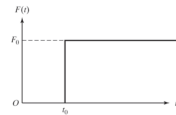
$$M\ddot{x} + K\dot{x} + Cx = F(t)$$

$$x(t) = \frac{1}{M\omega_d} \int_0^t f(\tau) e^{-\zeta\omega_n(t-\tau)} \sin[\omega_d(t-\tau)] d\tau$$

To use the integral, each different force needs its own integration

$$0 < t < t_0$$

$$x(t) = \frac{1}{M\omega_n} \int_0^t 0 \cdot \sin[\omega_n(t-\tau)] d\tau \rightarrow [x$$



$$t > t_0$$

$$x(t) = \frac{1}{M\omega_n} \int_0^{t_0} 0 \cdot \sin[\omega_n(t-\tau)] d\tau + \frac{1}{M\omega_n} \int_{t_0}^t F_0 \sin[\omega_n(t-\tau)] d\tau$$

$$x(t) = \frac{F_0}{k} (1 - \cos[\omega_n(t - t_0)])$$

Response to a General Periodic Force

The equation of motion would be

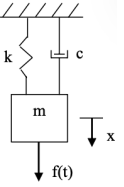
$$m\ddot{x} + c\dot{x} + kx = \frac{a_0}{2} + a_1 \cos(\omega_1 t) + b_1 \sin(\omega_1 t) + \dots$$

For multiple inputs you can split it up

The complete solution is

$$x_p(t) = \frac{a_0}{2k} + \sum_{n=1}^{\infty} \frac{a_n/k}{\sqrt{(1 - n^2 r^2)^2 + (2\zeta n r)^2}} \cos(n\omega_1 t - \phi_n) + \sum_{n=1}^{\infty} \frac{b_n/k}{\sqrt{(1 - n^2 r^2)^2 + (2\zeta n r)^2}} \sin(n\omega_1 t - \phi_n)$$

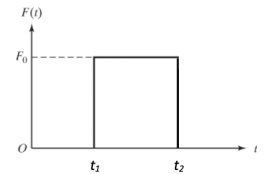
$$\text{Where } \phi_n = \tan^{-1} \left(\frac{2\zeta n r}{1 - n^2 r^2} \right)$$



$$\text{For } 0 < t < t_1$$

$$x(t) = \frac{1}{M\omega_n} \int_0^t 0 \cdot \sin[\omega_n(t-\tau)] d\tau$$

$$\rightarrow [x(t) = 0]$$



$$\text{For } t_1 < t < t_2$$

$$x(t) = \frac{1}{M\omega_n} \int_0^{t_1} 0 \cdot \sin[\omega_n(t-\tau)] d\tau + \frac{1}{M\omega_n} \int_{t_1}^t F_0 \sin[\omega_n(t-\tau)] d\tau$$

$$\rightarrow [x(t) = \frac{F_0}{k} (1 - \cos[\omega_n(t - t_1)])]$$

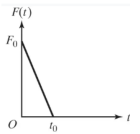
$$\text{For } t > t_2$$

$$x(t) = \frac{1}{M\omega_n} \int_0^{t_1} 0 \cdot \sin[\omega_n(t-\tau)] d\tau + \frac{1}{M\omega_n} \int_{t_1}^{t_2} F_0 \sin[\omega_n(t-\tau)] d\tau + \frac{1}{M\omega_n} \int_{t_2}^t 0 \cdot \sin[\omega_n(t-\tau)] d\tau$$

$$x(t) = \frac{F_0}{k} \{ \cos[\omega_n(t - t_2)] - \cos[\omega_n(t - t_1)] \}$$

$$\text{For } 0 < t < t_0$$

$$x(t) = \frac{1}{M\omega_n} \int_0^t F_0 \left(1 - \frac{\tau}{t_0} \right) \sin[\omega_n(t-\tau)] d\tau$$



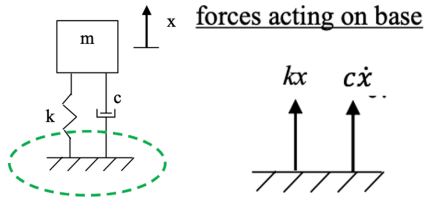
$$x(t) = \frac{F_0}{k} \left[1 - \frac{t}{t_0} - \cos(\omega_n t) + \frac{1}{\omega_n t_0} \sin(\omega_n t) \right]$$

$$\text{For } t > t_0$$

$$x(t) = \frac{1}{M\omega_n} \int_0^{t_0} F_0 \left(1 - \frac{\tau}{t_0} \right) \sin[\omega_n(t-\tau)] d\tau + \frac{1}{M\omega_n} \int_{t_0}^t 0 \cdot \sin[\omega_n(t-\tau)] d\tau$$

$$x(t) = \frac{F_0}{k\omega_n t_0} \{ [1 - \cos(\omega_n t_0)] \sin(\omega_n t) - [\omega_n t_0 - \sin(\omega_n t_0)] \cos(\omega_n t) \}$$

Vibration Isolation



Base citation is used

$$x(t) = X \sin(\omega t - \phi)$$

Maximum force to the ground is:

$$|F_T| = \sqrt{(kX)^2 + (c\omega X)^2}$$

$$F_{T,max} = X\sqrt{(k)^2 + (c\omega)^2} = kX\sqrt{1 + (2\zeta r)^2}$$

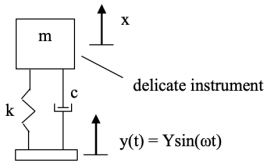
$$X = \frac{\frac{F_0}{k}}{\sqrt{(1 - r^2)^2 + (2\zeta r)^2}}$$

The transmissibility is given by:

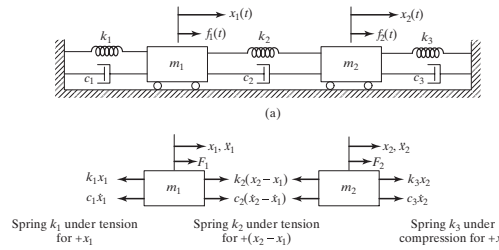
$$TR = \left| \frac{F_T}{F_0} \right| = \sqrt{\frac{1 + (2\zeta r)^2}{(1 - r^2)^2 + (2\zeta r)^2}}$$

Motion transmitted to mass from the base

$$TR = \left| \frac{X}{Y} \right| = \sqrt{\frac{1 + (2\zeta r)^2}{(1 - r^2)^2 + (2\zeta r)^2}}$$



Two Degrees of freedom systems



The equations of motion are as follows:

$$m_1 \ddot{x}_1 + (c_1 + c_2) \dot{x}_1 - c_2 \dot{x}_2 + (k_1 + k_2)x_1 - k_2 x_2 = f_1$$

$$m_2 \ddot{x}_2 - c_2 \dot{x}_1 + (c_2 + c_3) \dot{x}_2 - k_2 x_1 + (k_2 + k_3)x_2 = f_2$$

Coupling the equations, we can represent the system as matrices.

$$[m] \ddot{\vec{x}}(t) + [c] \dot{\vec{x}}(t) + [k] \vec{x}(t) = \vec{f}(t)$$

$$[m] = \begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix}, [c] = \begin{bmatrix} c_1 + c_2 & -c_2 \\ -c_2 & c_2 + c_3 \end{bmatrix}, [k] = \begin{bmatrix} k_1 + k_2 & -k_2 \\ -k_2 & k_2 + k_3 \end{bmatrix}$$

$$\vec{x}(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}, \dot{\vec{x}}(t) = \begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix}, \vec{f}(t) = \begin{bmatrix} f_1(t) \\ f_2(t) \end{bmatrix}$$

Undamped system which give the following solution no damping

$$\vec{X} = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}, x_1(t) = X_1 \cos(\omega t), x_2(t) = X_2 \cos(\omega t)$$

$$x_p(t) = \vec{X} \cos(\omega t), \ddot{x}_p(t) = -\omega^2 \vec{X} \cos(\omega t)$$

This is a good way to find omega: $DET(\vec{K} - \omega^2 \vec{M}) = 0$

To find the mode, plug in omega: $(\vec{K} - \omega^2 \vec{M}) \vec{X} = \vec{0}$

$$\text{Solve for } r_1 = \frac{x_2^{(1)}}{x_1^{(1)}} \text{ Modes are of form: } \vec{X}^{(1)} = \begin{Bmatrix} X_1^{(1)} \\ X_2^{(1)} \end{Bmatrix} = \begin{Bmatrix} X_1^{(1)} \\ r_1 X_1^{(1)} \end{Bmatrix} = \begin{Bmatrix} 1 \\ r_1 \end{Bmatrix}$$

$$r_2 = \frac{x_2^{(2)}}{x_1^{(2)}} \quad \vec{X}^{(2)} = \begin{Bmatrix} X_1^{(2)} \\ X_2^{(2)} \end{Bmatrix} = \begin{Bmatrix} X_1^{(2)} \\ r_2 X_1^{(2)} \end{Bmatrix} = \begin{Bmatrix} 1 \\ r_2 \end{Bmatrix}$$

Matrix Math

$$\vec{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\rightarrow DET(\vec{A}) = ad - cb$$

$$\vec{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\rightarrow ADJ(\vec{A}) = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Two DOF Undamped Forced Vibration

$$\vec{M} \ddot{\vec{x}} + \vec{K} \vec{x} = \vec{f} \sin(\omega t)$$

Assume solution for steady state:

$$\vec{x}_{ss}(t) = \vec{X} \sin(\omega t)$$

Where:

$$\vec{X} = \frac{ADJ(\vec{Z})}{DET(\vec{Z})} \vec{f}$$

$$\vec{x} = \frac{ADJ(\vec{Z})}{DET(\vec{Z})} \vec{f} \sin(\omega t)$$

$$\text{Where: } \vec{Z} = \vec{K} - \omega^2 \vec{M}$$

Two DOF Damped Forced Vibration

$$\vec{M} \ddot{\vec{x}} + \vec{C} \dot{\vec{x}} + \vec{K} \vec{x} = \vec{f} \sin(\omega t)$$

To simplify, apply Euler formula: $e^{i\phi} = \cos(\phi) + i\sin(\phi)$

$$\vec{M} \ddot{\vec{x}} + \vec{C} \dot{\vec{x}} + \vec{K} \vec{x} = \vec{f} e^{i\omega t}$$

Form of solution is:

$$\vec{x}(t) = \vec{X} e^{i\omega t}, \quad \vec{X} = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}, \quad \vec{X} = \frac{ADJ(\vec{Z})}{DET(\vec{Z})} \vec{f}$$

Where the impedance matrix is:

$$\vec{Z} = \vec{K} - \omega^2 \vec{M} + i\omega \vec{C}$$

$$\text{The solution is: } \vec{x}(t) = \text{Im} \left[\frac{ADJ(\vec{Z})}{DET(\vec{Z})} \vec{f} e^{i\omega t} \right], \quad \vec{X} = \text{Im} \left[\frac{ADJ(\vec{Z})}{DET(\vec{Z})} \vec{f} \right]$$

If the forcing function is $\vec{f} \cos(\omega t)$, the solution is just the real part: $\vec{M} \ddot{\vec{x}} + \vec{C} \dot{\vec{x}} + \vec{K} \vec{x} = \vec{f} \cos(\omega t)$

Vibration Absorbers

Natural frequencies: Ω_1 & Ω_2

Damping frequencies: ω_1 & ω_2

$$r_1 = \frac{\Omega_1}{\omega}, \quad r_2 = \frac{\Omega_2}{\omega}$$

We can also solve for the two r values using:

$$r_1^2 = \left(1 + \frac{\mu}{2}\right) - \sqrt{\left(1 + \frac{\mu}{2}\right)^2 - 1}$$

$$r_2^2 = \left(1 + \frac{\mu}{2}\right) + \sqrt{\left(1 + \frac{\mu}{2}\right)^2 - 1}$$

$$\text{Where: } \mu = \frac{m_2}{m_1}$$

Tuning the Absorber: $k_2 = m_2 \omega^2$

$$\text{How far will it move: } X_2 = -\frac{f_0}{k_2}$$